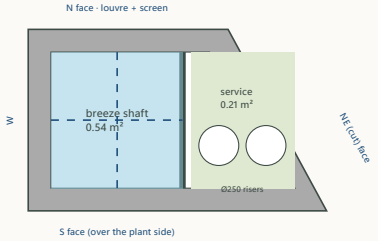
Air system study v3 · one tower, two separate air paths · Taif / Palm House V7.1

Owner goals 1 Nov 2026 (design review E-01) · a gentle natural breeze for the courtyard seating in pleasant weather, and the fresh-air intake of the house's ventilation unit all year · no water in the air system, no pads, no earth pipes, no human entry · **every size and performance figure is ASSUMED, to be designed by the MEP engineer** compiling this study is not engineering approval

1 · The concept

- **One tower** in the tip of the cut corner (1.37 m² on walls, **0.83 m² clear**, assumed), the symbol of the house, rising 2.5 m above the roof finish (head +7.3 … +9.8), the one element above the attic (`studies/roof.typ`).
- **Path (a), the courtyard breeze:** a four-sided head → a breeze shaft → a damper → a shallow channel in the ground slab → the grille under the outdoor TV: a gentle natural breeze for the seating in pleasant weather; not cooling (if it is too hot the family does not sit there); no water, no pads, no earth pipe, no fan (a small assist fan is **proposed**, § 4).
- **Path (b), the house's fresh air:** an airtight service compartment in the same tower with its own louvre and an accessible pre-filter, feeding the ventilation unit in a small plant room on the roof next to the tower; its supply and extract risers run down the same compartment to both floors' ceilings. The two paths never meet.
- **Low maintenance, no entry:** every routine task is done from the roof (the head, the flaps, the damper, the pre-filter), from the plant room (the unit, the filters, the core), from the laundry and kitchen floors (the channel's covers) or from the courtyard (the grille). No shaft, pit or buried space is entered (§ 7).



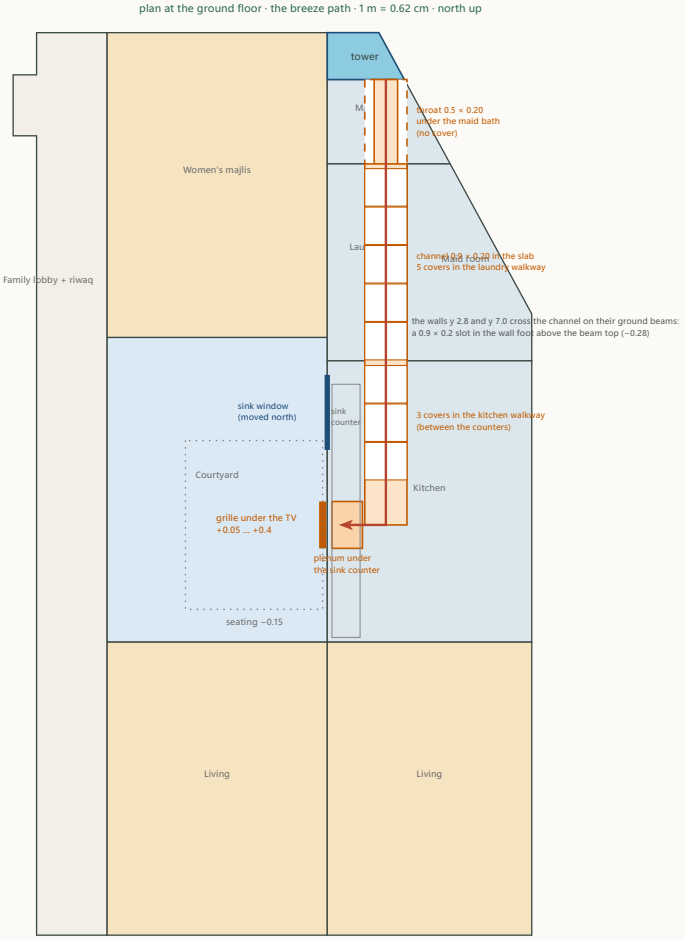
tower plan at the head · the + divider (dashed) only in the head, each quarter with a back-draught flap · 1 m = 2.4 cm

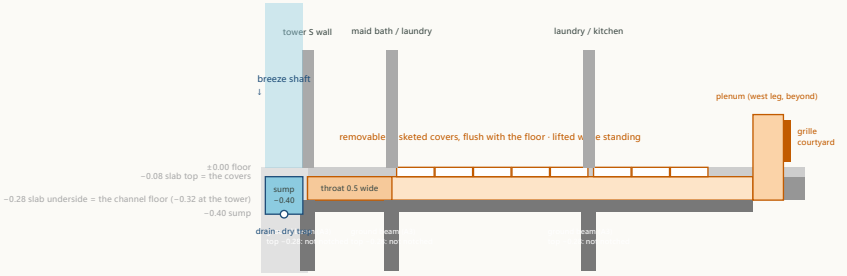
2 · The tower

- **Clear section 0.83 m²** (assumed, inside the walls), split by an airtight partition (0.1 m, block or a sealed steel sheet on a frame) into the **breeze shaft 0.54 m²** (west) and the **service compartment 0.21 m²** (east): it holds both, **to confirm with the real wall thicknesses** if the engineer thickens the walls, the smallest enlargement is 0.25 m on the cut side (a proposal only, not applied).
- **Head (Yazd badgir type), four faces, in the house's style:** louvred openings +8.2 … +9.4 on all four faces (3.84 m² gross), honey-stone surrounds, timber louvres. **The problem checked:** an undivided head open on four sides short-circuits (the wind enters windward and leaves leeward without reaching the courtyard). **The simplest arrangement that works:** the head only is divided into four quarters by a thin + divider (1.2 m high); at the foot of each quarter a **lightweight back-draught flap** (a gravity-hinged aluminium or fabric flap) closes under the leeward suction and opens under the windward pressure, so only the windward quarter(s) feed the single shaft below; no X dividers in the shaft, no wind sensor, no motor. **The flaps' free opening and their damping are the MEP engineer's.**
- **Louvres and screens, cleanable from the roof:** on each face a weather louvre (≈ 50 % free), a 20 mm bird mesh and a 2 mm insect mesh (≈ 60 % free; a 1.2 mm mosquito mesh halves the free area: the family decides), each on a hinged frame opened from the roof strip beside the tower (a 2 m step ladder) and brushed / vacuumed. **Their pressure loss counts for little** because the head's open area (1.15 m² net) is ten times the channel's throat: the losses sit in the channel and the grille (§ 4). No fine filter on the breeze path: natural wind cannot push through one, and the air goes outdoors.
- **Damper:** a motorised blade damper across the breeze shaft near the TOP (just under the flaps, inside the tower at ≈ +8.0), a standard **spring-return HVAC actuator** (closes on power loss; ≈ 5 W holding, 24 V), on a wall switch beside the seating (by the TV) with an "open" light; the actuator inside the tower behind a hinged panel, reached from the **roof hatch** in the tower's south face (+7.6 … +8.4) by a hand from the roof: no entry.
- **ERV intake compartment:** the service compartment's top 1.5 m (+7.3 … +8.8) is an airtight box with its own louvre on the north face (0.8 × 0.5 at +8.4 … +8.9, a bird mesh, no fine screen) and a slide-in **pre-filter** (ISO Coarse 60 %, 500 × 500) behind it, pulled out through the same roof hatch (a second panel); below the box the two Ø250 risers; a Ø315 insulated duct leaves the box through the tower's south wall into a low plastered duct box (0.4 high, below the cornice coping) to the plant room 1.9 m south.

Air system study v3 · checks passed (8)

3 · The courtyard breeze path: the channel in the slab, its covers, the plenum and the grille · plan and long section





long section along the channel (x 10.2), looking east, heights × 2.5 · 1 m = 0.62 cm · the channel never leaves the slab depth

- **Route:** from the breeze shaft's foot (the tower's south wall, y 1.0) **straight south on x 10.2** under the maid bath (y 1.0–2.8), the laundry walkway (2.8–7.0) and the kitchen walkway between the counters (7.0–10.5), then **one 90° bend west** under the sink counter (y 10.0–11.0) into the plenum in the wall base and the grille: axis 10.5 m, one bend. The slot through each of the three walls (y 1.0, 2.8, 7.0) is 0.9 × 0.20 in the wall foot **above the ground beam's top (−0.28)**: no beam, footing or floor is cut or lowered.
- **Size and levels:** clear 0.9 × 0.2 m (0.18 m²) from the slab top (−0.08, the covers flush with the finished floor) to the slab underside (−0.28), its own base slab 0.10 below; under the maid bath a **throat 0.5 × 0.2** between the shower and the WC (the WC moved to the cut-wall corner, `studies/east_wing.typ`): 0.1 m², the controlling section. The floor falls 0.5 % back to the tower's **sump** (−0.40, a Ø75 drain with a waterless dry trap against odours, to the east yard): rain that passes the louvres and any condensation leave at the tower, never towards the courtyard.
- **Access without entry:** 8 removable sealed covers (0.9 × 0.8, stainless trays stone-faced like the floor, gasketed against wash-water and insects, lifted with two suction cups) in the laundry and kitchen walkways; the throat section (1.8 m) and the sump are reached with a flat brush and a vacuum rod from the first cover and from the shaft's foot; the plenum and the west leg (0.65 m) from the courtyard with the grille removed. The channel's surfaces smooth (a sealed epoxy render), the corners coved.
- **Outlet:** the decorative grille under the TV, 1 × 0.35 m at +0.05 … +0.4 (0.2 above the seating floor), with a **manually adjustable louvre** that directs the air along the floor to the benches, clear of the cushions (no bench on the east side) and 1.6 m from the fountain enclosure's splash; the grille lifts off for cleaning.

4 · The breeze, estimated honestly · 5 · The house ventilation (the sealed house)

4 · What the courtyard will feel

The path's resistance, referred to the throat's velocity (**assumed loss coefficients, ASHRAE Fundamentals ch. 21 orders of magnitude**):

Element	ζ at the throat
head louvres + screens (4 faces, 3.84 m² gross, 30 % free → 1.15 m²)	0.02
back-draught flaps and the shaft (0.54 m²)	0.03
damper, open (in the shaft)	0.02
the throat under the maid bath (contraction + expansion)	1.5
channel friction, 10 m, 0.18 m²	0.2
the bend and the plenum	0.25
the grille (0.35 m², 50 % free)	0.82
Total	**2.83**

Driving pressure: wind at the head ½ρv²·ΔCp with ρ 1.15 kg/m³ (Taif, 1 700 m) and ΔCp 0.8 (windward face against the courtyard well, assumed); in still air the stack of an 8 m tower with 3 K difference: 0.9 Pa.

Wind at the head	Δp	flow	grille	felt at the benches	What it is
weak, 3 m/s	4.1 Pa	0.16 m³/s (574 m³/h)	0.91 m/s	≈ 0.26 m/s	barely perceptible: a faint stirring at the floor
moderate, 6 m/s	16.6 Pa	0.32 m³/s (1148 m³/h)	1.82 m/s	≈ 0.51 m/s	a gentle, pleasant breeze on the legs and hands
still (stack only)	0.9 Pa	0.07 m³/s	0.43 m/s	≈ 0.12 m/s	nothing felt

- **Honest conclusion:** the breeze is perceptible only with a **moderate wind at the head** (≈ 5 m/s and more at 9 m above the ground); Taif's spring and summer afternoons often have it, calm evenings do not. In still weather the tower gives nothing the family will notice. The path is as short and open as the house allows (the throat under the maid bath halves what a 0.18 m² channel would pass); the head's screens are not the limit.
- **Proposal, not added:** a **small quiet assist fan** in the breeze shaft's foot (an EC inline fan ≈ Ø315, 0.15–0.25 m³/s at ≤ 35 dB(A) in the courtyard, 40–60 W, on the same wall switch as the damper, reached from the first laundry cover): it gives the moderate-wind breeze on any evening. **The MEP engineer sizes it or confirms the natural path alone.** Wind data for Taif (the airport station) to be checked; the figures above are a concept estimate, not a design.

5 · The house's ventilation

- **One unit** for the whole house in a **small weather-protected plant room on the roof** next to the tower (`studies/roof.typ` Z2a): x 9.1–10.6, y 2.9–5.4, inside the attic line; 200 mm walls, 1.2 × 2.2 m clear, 2 m clear height (roof top +9.42, 0.32 m above the attic coping, like the stair room); a 0.8 m door on its west side facing the roof walkway from the stair room; the unit floor-standing on a plinth with 0.8 m in front for the filters and the core pull, 0.5 m at the sides. **Unit (assumed):** a 750 m³/h class counterflow HRV / ERV (sensible core; enthalpy core optional), floor-standing on a plinth in the plant room: 750 m³/h class, ≈ 1.2 × 0.7 × 0.7 m, ≈ 110 kg, 90 W at the normal rate and ≈ 350 W at boost. **Footprint, weight and power per the chosen unit.**
- **ERV or HRV:** Taif is dry and high (summer 30–35 °C by day with low humidity, winter nights near 5 °C). Moisture recovery gives little here: in summer the outdoor air is dry anyway, in winter it only slows the drying of the indoor air. A **sensible-only HRV** (or a unit whose enthalpy core can be swapped for a sensible one) is the better fit: simpler core, no condensate in summer, the winter heat recovery is what matters. **The MEP engineer decides with the unit's data.**
- **Preliminary airflow** (ASHRAE 62.2-2022 §4.1.1: 0.15 L/s·m² of floor + 3.5 L/s per bedroom + 1): conditioned floor ≈ 471 m² clear (512 m² on axes), 5 bedrooms → **92 L/s = 330 m³/h** continuous. **Guest gatherings:** 24 men in the majlis (47 m²) at the ASHRAE 62.1 assembly rate (3.8 L/s·person + 0.3 L/s·m²) add 105 L/s → **boost 197 L/s = 708 m³/h**. The unit above covers it ✓. **Rates to be confirmed against SBC 501 / 1101 and the owner's occupancy.**
- **Intake and exhaust:** the intake from the tower's service compartment (north face at the head); the exhaust on the plant room's **east** wall, blowing east over the condenser zone; distances below.
- **Filters:** a coarse pre-filter (ISO Coarse 60 %) in the tower compartment and **one fine filter** at the unit's supply, **provisionally ISO ePM1 ≥ 50 %** (**to confirm with the equipment**); a differential-pressure switch or the unit's own filter timer as the service indicator, shown at the plant room door and on the panel.
- **Supply and extract:** supply to every bedroom (1–3, the kids' room, the suite), the living, the family living, the office and the men's majlis (with the boost); extract from every bathroom and WC (the family bath, bath 3, the suite bath and WC, the family WC, the guest WC, the maid bath) and the laundry; transfer air from the rooms to the extract points through **door undercuts of 15–20 mm** or transfer grilles over the bathroom doors, so the system works with the bedroom doors closed.
- **The kitchen hood stays separate:** 750 m³/h out through the east wall; **make-up air** through a filtered, motorised inlet (≈ 300 × 300) high on the kitchen's east wall north of the window, interlocked with the hood, so the heavy daily cooking never depressurises the sealed house (a 0.9 m³/h gap would otherwise pull dust through every crack). **The MEP engineer sizes it.**

Intake / exhaust separations (from the plan; the basis in the last column)

Pair	distance	allowance	basis
ERV intake ↔ ERV exhaust	4.1 m	3 m	ASHRAE 62.1-2022 Table 5-1: class 2 exhaust (toilets) → intake ≥ 3 m (10 ft); the exhaust blows east, the intake faces north
ERV intake ↔ kitchen hood outlet (east wall)	12.8 m	4.6 m	class 3 (cooking) → ≥ 4.6 m (15 ft); the hood outlet is 6.6 m below the intake
ERV intake ↔ plumbing vent stack	8.7 m	3 m	vents → ≥ 3 m (10 ft), and the intake 0.9 m above the vent top
ERV intake ↔ LPG cabinet (yard)	12.7 m	4.6 m	a leak source at ground level: class 3 allowance
ERV intake ↔ generator point (yard)	15.3 m	9 m	class 4 (engine exhaust) → ≥ 9 m (30 ft); the generator's exhaust pipe led away from the house
ERV intake ↔ nearest AC condenser (Z2b)	5.2 m	2 m	not an exhaust: 2 m from the hot plume (manufacturer); the intake is 1 m above the units
ERV exhaust ↔ nearest AC condenser	1.2 m	1 m	stale air into a condenser is harmless; keep 1 m for service

ASHRAE 62.1-2022 §5.5.1 and Table 5-1 (the 3 m / 4.6 m / 9 m figures are the IFC-era "starting allowances" the SBC 501 follows, **not verified**); the gas cabinet and the generator point are site-plan proposals and may move; proposal: keep the condenser zone south of y 5.6 so the intake stays upwind of their plumes in the north-west wind.

Ducts and ceiling heights

- **Vertical route:** from the plant room through the duct box into the tower's service compartment: two Ø250 risers (supply, extract) down to the first-floor ceiling zone (+6.85) through the tower's south wall into bedroom 1's ceiling and on to the gallery, and to the ground-floor zone (+3.0 … +3.3) into the maid bath / laundry ceiling and on to the kitchen, the dining, the riwaq and the two majlises. Flat ovals 100 × 300 where the zone is thin, Ø160 branches, acoustic attenuators at the plant room. **Sizes assumed.**

Zone	soffit	duct	ceiling
GF rooms under the 0.20 slabs (women's majlis, riwaq, the wash block, the maid wing)	+3.32	0.1	**+3.17**
GF rooms under the 0.30 ribbed bays (kitchen, dining, men's majlis)	+3.22	0.1	**+3.07**
under beam B1 (women's majlis / kids' wall, soffit +3.20): a 0.3 m band	+3.2	—	**+3.05**
under beam B4 (the riwaq's east wall line, soffit +3.10): a band over the riwaq	+3.1	—	**+2.95**
under beam B5 (the living / dining ceiling step, soffit +3.05): a 0.3 m band at the dining's west edge	+3.05	—	**+2.9**
kitchen (the hood duct 0.25 + the extract duct in the same zone)	+3.22	0.25	**+2.92**
FF rooms under the 0.20 roof slab (bedrooms 2 / 3, the kids' room, the family living, the gallery)	+7	0.1	**+6.85**
FF rooms under the 0.30 roof bays (the suite, the office, the gym)	+6.9	0.1	**+6.75**

- **Finished clear heights:** ≥ +3.07 in the ground-floor rooms, +2.90 only in the 0.3 m band under B5 at the dining's west edge (the living has no ceiling); the first floor ≥ +6.75 (3.15 m clear). **No room gets too low**; the lowest band is at the ceiling step the living already has.

6 · Controls · 7 · Maintenance (no routine task enters a shaft, pit or buried space) · 8 · Decisions and engineering checks still open

6 · Controls

- **Continuous quiet running** at the normal rate (330 m^3/h), **boost** (708) from the kitchen and bathroom switches (timed) and a "guests" button in the hospitality kitchen for the majlis evenings; the fan speeds set at commissioning.
- **The courtyard damper is independent:** its own switch by the seating; spring-return, so it **closes on power loss** and when the switch is off; the assist fan (if added) on the same switch.
- **Dust-event mode** (a switch at the family entrance or a weather-station input): the unit drops to its **minimum balanced** rate (supply = extract) so the house is never depressurised, the kitchen hood's make-up inlet stays interlocked, the courtyard damper closes; the unit is **not** assumed to recirculate: indoor air cleaning during a storm is **separate equipment** (portable HEPA units in the bedrooms and the living, a proposal).
- **Power failure:** the unit stops, the damper springs shut, the sealed house stays sealed; the alarms keep running on their batteries (`studies/fire_escape.typ`); on restoration the unit restarts at its last setting. The unit may sit on the essentials circuit if the electrical designer agrees (≈ 100–350 W).
- **Indicators:** the filter-service light at the plant room door and on the panel; the damper "open" light at the seating switch.

8 · Decisions (owner) and engineering checks still open

- **Owner:** the assist fan (yes / no); the mosquito mesh (1.2 mm, more loss) or the fly mesh (2 mm); the plant room's visibility (0.3 m above the attic coping, like the stair room); the make-up air inlet's position on the kitchen's east wall; the condensers kept south of y 5.6 (proposal).
- **MEP engineer:** every rate, size and pressure drop above; the flaps and the damper; the HRV / ERV choice and the unit; the duct sizes and the attenuators; the make-up air; the sump drain and its trap; the controls; the Taif wind data for the breeze.
- **Structural engineer:** the channel in the slab (0.9 × 0.2 within the slab depth, its base slab, the three wall slots above the ground beams), the tower's partition and the roof penetrations, the plant room's walls and the unit's load (`studies/structure.typ` X19).
- **3D and other files:** `3d/build_3d.py` / `check_3d.py` still build and check the v2 pit, chamber and duct: to be rewritten for v3 at the next 3D build (`docs/PENDING_CHANGES.md`).

7 · Maintenance (initial intervals; adjust to the manufacturer and the dust)

Component	Task	Reached from	Interval	If neglected
Head louvres, bird and insect screens (4 faces)	brush / vacuum the dust; check the mesh	the roof strip beside the tower, hinged frames, a 2 m step ladder	3 months; after every dust storm	the breeze drops; insects and birds enter the shaft
Back-draught flaps (4)	check they swing free, clean the seats	the roof hatch in the tower's south face, by hand	12 months	a leeward quarter stays open: the head short-circuits, no breeze
Courtyard damper and its actuator	cycle it, check the spring return and the switch	the roof hatch, a hinged panel	12 months	the damper sticks open (dust and cold air into the courtyard) or closed
ERV pre-filter (tower compartment)	slide out, replace	the roof hatch, a second panel	3 months	the unit's fine filter clogs early; less fresh air
Unit filters (plant room)	replace the fine supply filter and the extract filter	the plant room, the front panel	3–6 months (the indicator)	less air, more fan power, dust bypass
Heat-recovery core	pull, wash, dry	the plant room, 0.8 m clear in front	12 months	lost recovery, odours
Exhaust louvre and the make-up air inlet	clean the mesh; test the motorised inlet with the hood	the roof (east wall of the plant room) / the kitchen wall	6 months	the hood depressurises the house; dust in
Channel covers and channel	lift the trays, vacuum the channel, check the gaskets	the laundry and kitchen walkways, standing	6 months; after a storm	dust blown into the courtyard; wash-water into the channel
Throat section and sump	brush with a rod; check the dry trap, pour water once a year	the first laundry cover and the shaft's foot (no entry)	6 months	odours from the drain; a blocked sump
Plenum and grille	lift off the grille, vacuum the plenum, set the louvre	the courtyard	1 month (dust)	dust on the seating; a blocked grille
Supply and extract grilles in the rooms	vacuum	the rooms, a step ladder	6 months	noise, less air
Fountain pump and filter (separate system)	see `studies/courtyard_layout.typ`	the enclosure's cover	per the pump	—

Every task is done from the roof, the plant room, a room's floor or the courtyard. Nothing requires entering the tower, the channel or any pit: the sump is 0.32 m deep and reached by hand and rod.